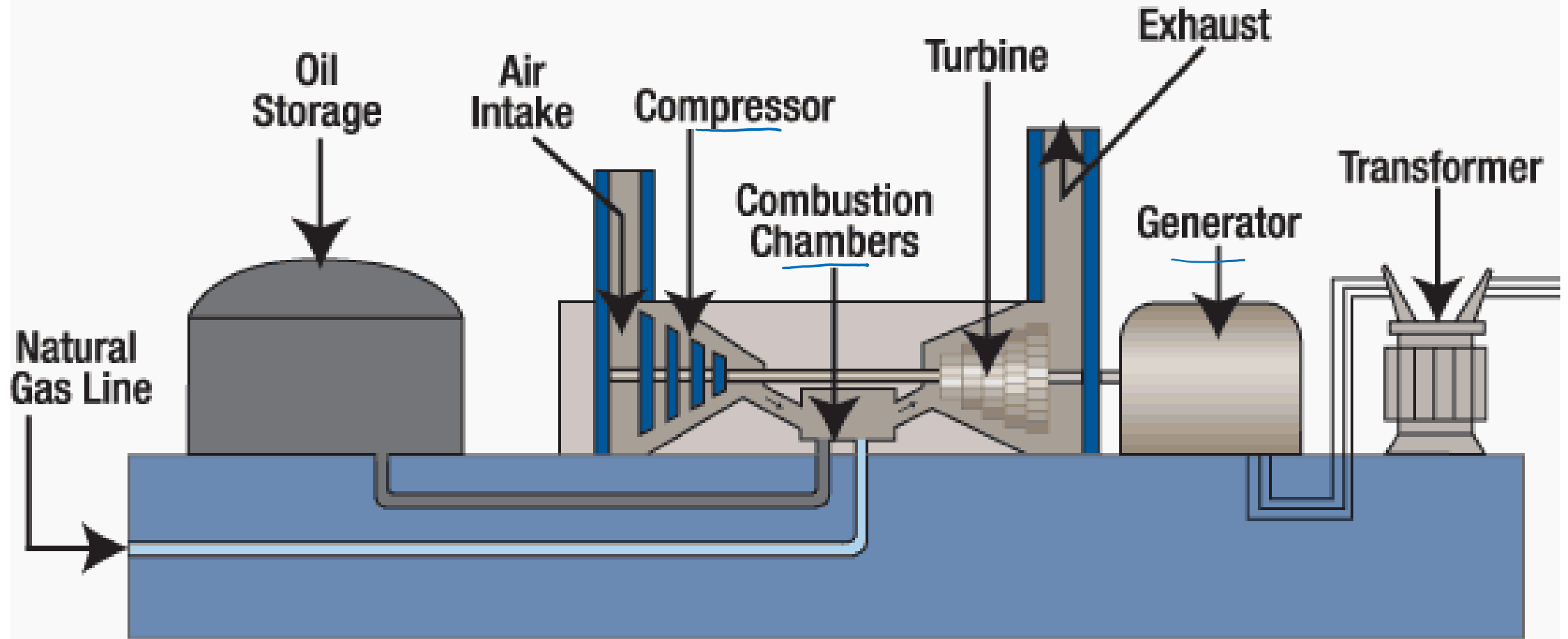




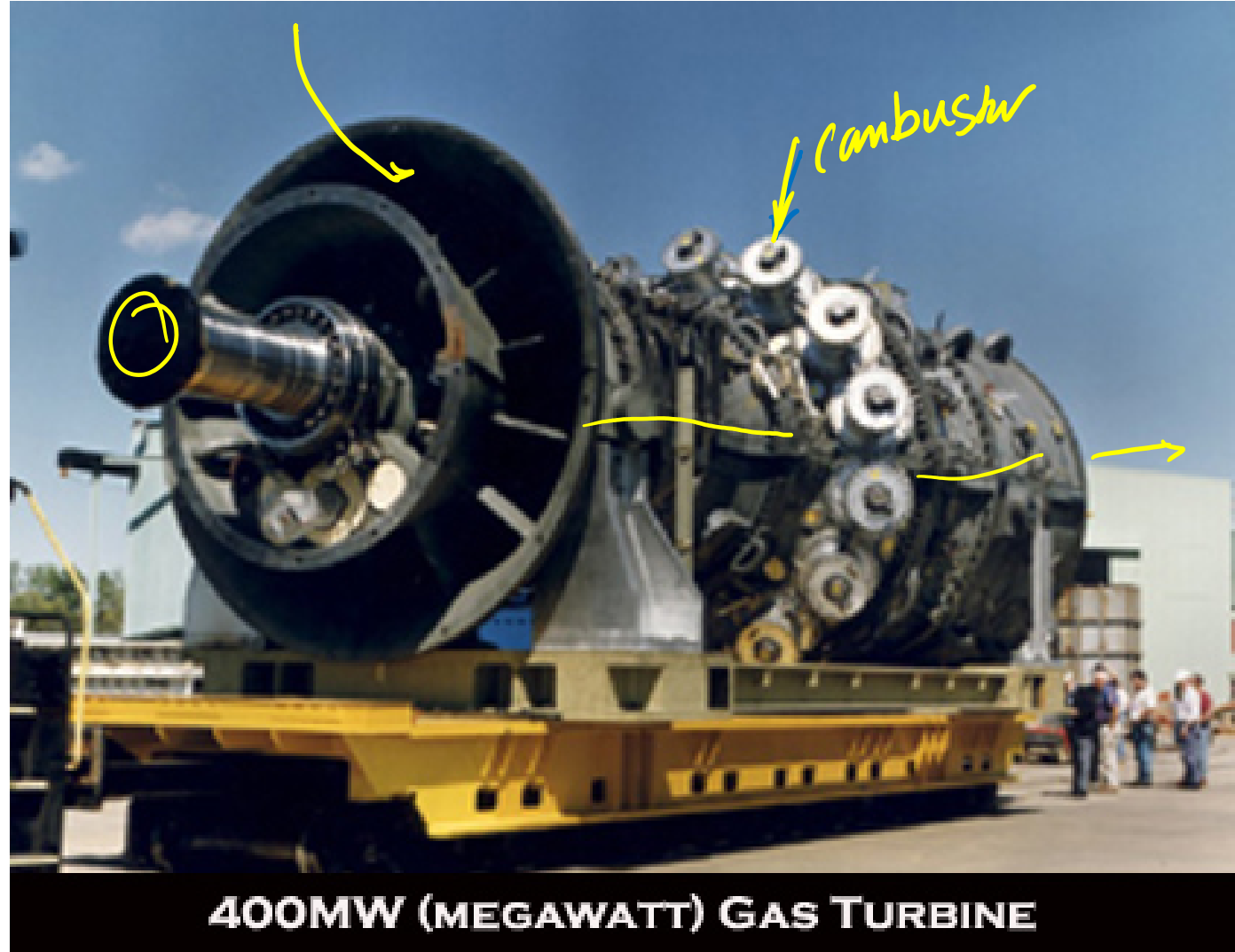
ME 300: Engineering Thermodynamics

Brayton Cycle

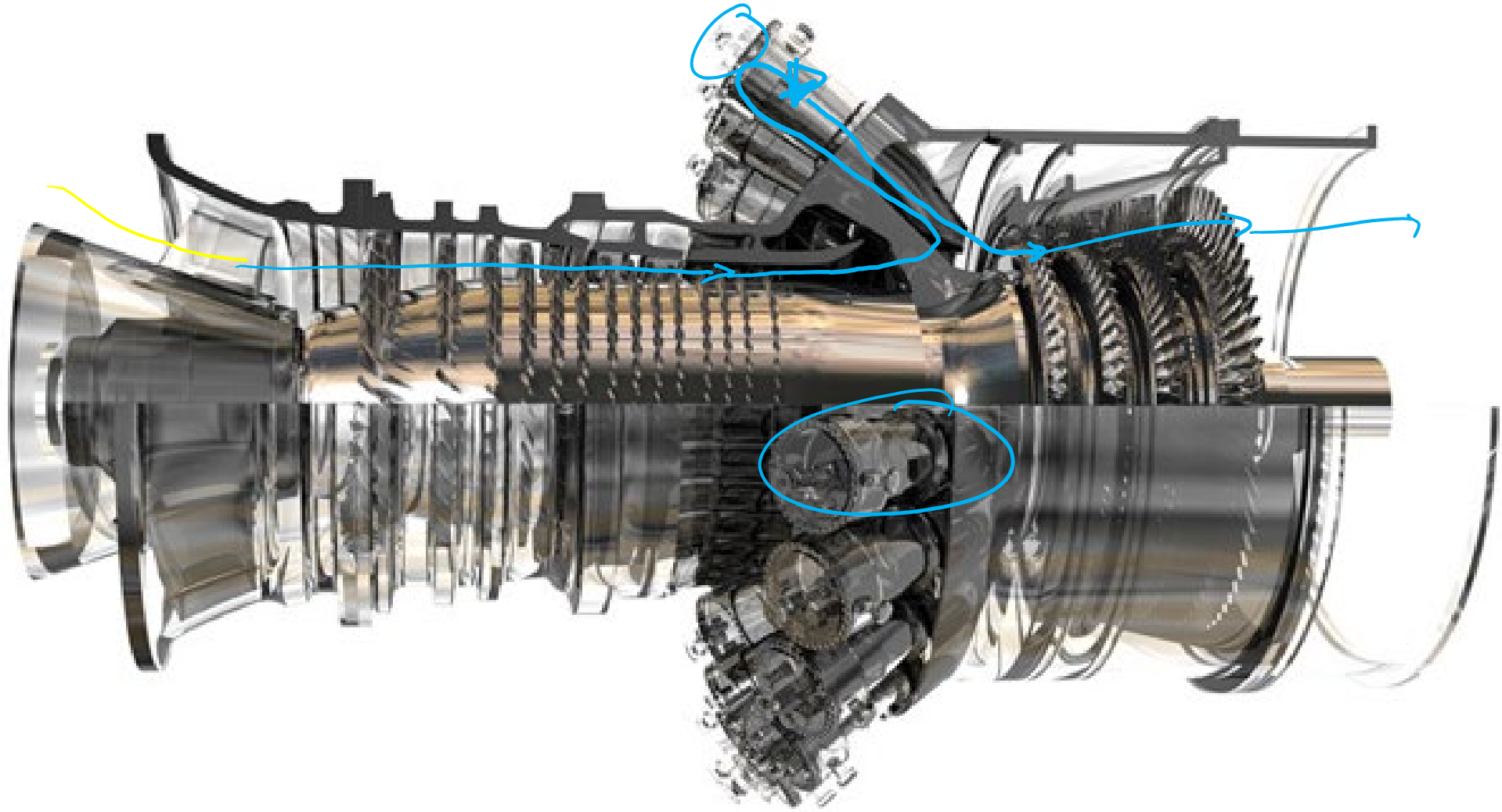
Power generation gas turbines



Power generation gas turbines

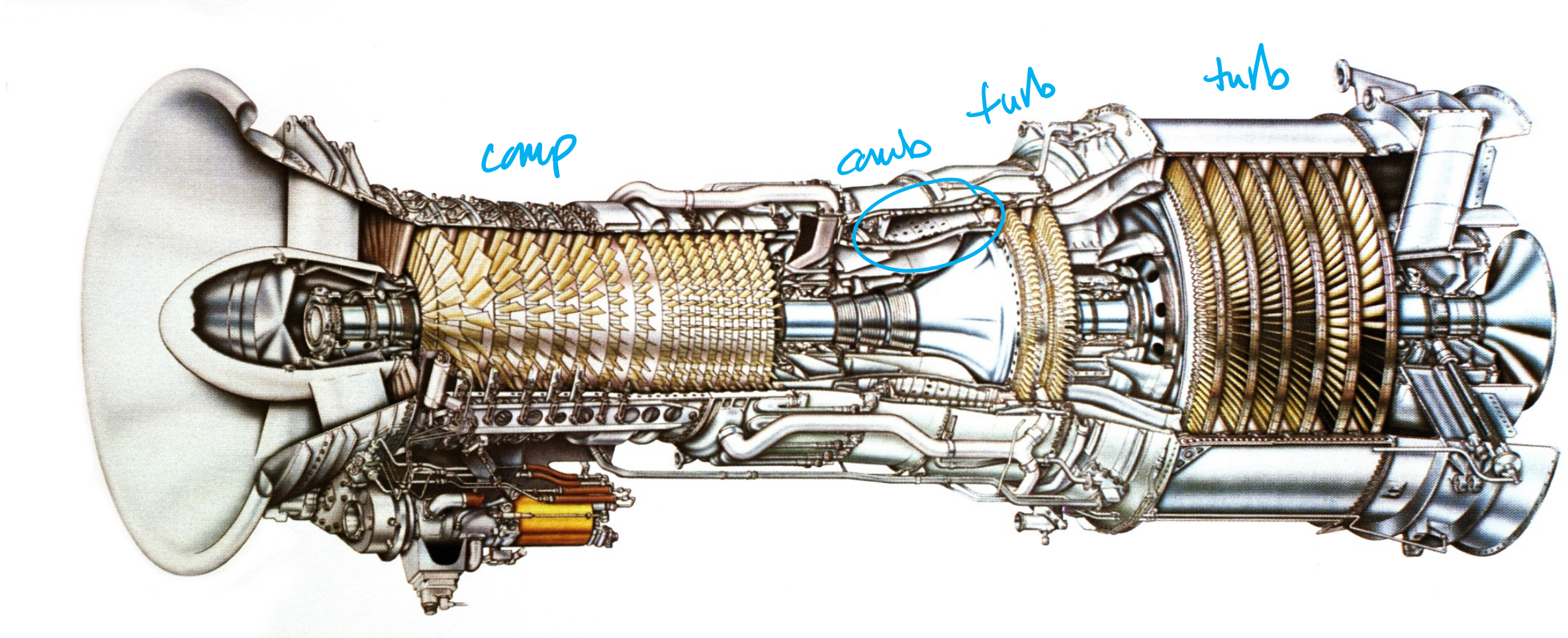


Power generation gas turbines



“Heavy Frame”

Power generation gas turbines



“Aeroderivative”

Brayton cycle - ideal

Air standard

↳ air working fluid

↳ air ideal gas

→ γ, c_p const, $\gamma = \frac{c_p}{c_v}$

↳ treat cycle as closed

↳ combustion = heat addition

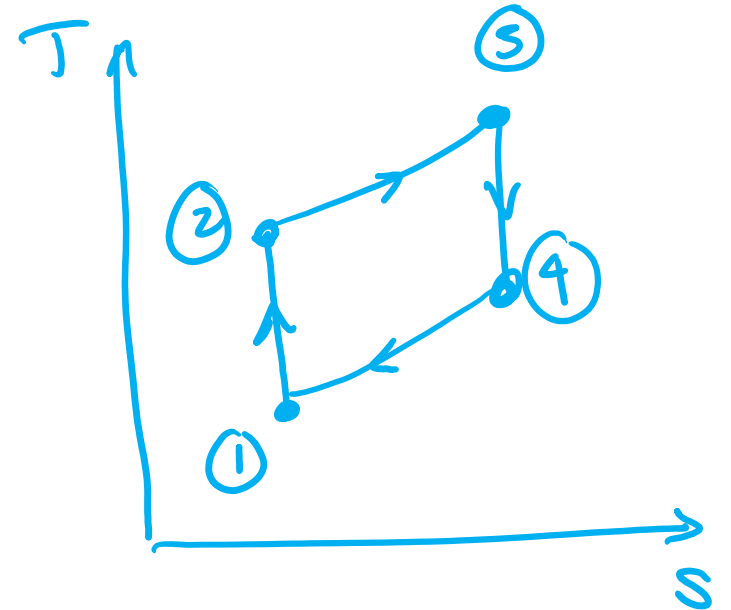
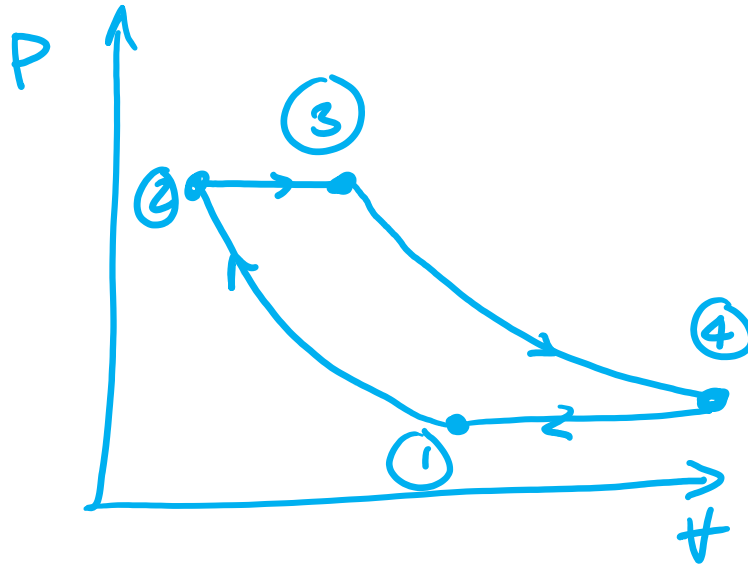
$\frac{P_2}{P_1}$ = overall pressure ratio "OPR"

1-2: isentropic comp. - compressor

2-3: isobaric heat in - combustor

3-4: isentropic expansion - turbine

4-1: isobaric heat rejection - ???



Cycle analysis

$$\dot{m} (\Delta h + \Delta ke + \Delta pe) = \dot{Q} - \dot{W}$$

1-2 : compressor

$$\dot{m} (h_2 - h_1) = -\dot{W}_2 \rightarrow \dot{m} c_p (T_2 - T_1) = -\dot{W}_2$$

2-3 : combustor

$$\dot{m} (h_3 - h_2) = {}_2\dot{Q}_3 \rightarrow \dot{m} c_p (T_3 - T_2) = {}_2\dot{Q}_3$$

3-4 : turbine

$$\dot{m} (h_4 - h_3) = -{}_3\dot{W}_4 \quad$$

4-1 :

$$\dot{m} (h_1 - h_4) = {}_4\dot{Q}_1$$

Efficiency

$$\eta_{th} = \frac{\dot{W}_{net}}{\dot{Q}_{in}} = \frac{\overset{<0}{\dot{W}_2} + \overset{>0}{\dot{W}_4}}{\dot{Q}_3} = \frac{(T_2 - T_1) + (T_4 - T_3)}{T_3 - T_2}$$

$$\rightarrow T_1 P_1^{\frac{1-\gamma}{\gamma}} = T_2 P_2^{\frac{1-\gamma}{\gamma}}$$

$$\eta_{th} = 1 - \left(\frac{P_2}{P_1} \right)^{\frac{1-\gamma}{\gamma}}$$

OPR $\uparrow$, $\eta_{th} \uparrow$